ECG Signal Classification and Heart Disease Prediction Using Hybrid CNN and RNN Model

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***Abstract*—Electrocardiogram (ECG) analysis is an important tool for identifying cardiac abnormalities, but manual interpretation of large-scale 12-lead ECG recordings can be time-consuming and requires significant clinical expertise. This paper presents a hybrid deep learning approach for automatic multi-label classification of 12-lead ECG signals using the PTB-XL dataset. The proposed framework performs signal preprocessing through resampling at 200 Hz, Butterworth band-pass filtering from 0.5 to 40 Hz, per-lead normalization, and fixed-length segmentation of 10 s recordings. A hybrid neural network is then developed by combining a multi-kernel one-dimensional convolutional neural network (CNN), residual learning, and squeeze-and-excitation attention for morphological feature extraction with bidirectional long short-term memory (Bi-LSTM) and bidirectional gated recurrent unit (Bi-GRU) networks for temporal feature modeling. The extracted representations are fused and used to predict multiple diagnostic categories simultaneously through independent sigmoid outputs. Binary cross-entropy with logits is employed as the training objective. The proposed framework is designed to capture both local waveform characteristics and long-range temporal dependencies present in ECG signals. Model performance is assessed using macro area under the receiver operating characteristic curve (AUC), precision, recall, F1-score, per-label accuracy, exact-match accuracy, and confusion matrices. The developed approach provides an end-to-end framework for automated multi-label ECG abnormality classification and can support computer-aided cardiac assessment.**

Index Terms— Electrocardiogram (ECG), multi-label classification, deep learning, convolutional neural network (CNN), bidirectional long short-term memory (Bi-LSTM), bidirectional gated recurrent unit (Bi-GRU), PTB-XL dataset, squeeze-and-excitation attention, biomedical signal processing.

I. Introduction

Electrocardiography (ECG) is a widely used non-invasive technique for recording the electrical activity of the heart. A standard 12-lead ECG provides information about cardiac rhythm and electrical activity from multiple viewpoints and is commonly used for identifying abnormalities such as atrial fibrillation, conduction disorders, and other cardiac conditions. However, accurate interpretation of ECG recordings requires considerable clinical expertise and can become difficult when large numbers of recordings must be analyzed. These challenges have motivated the development of computer-aided ECG analysis systems that can assist clinicians by automatically identifying relevant cardiac abnormalities.

Recent advances in deep learning have demonstrated the potential of neural networks for automatic ECG analysis. Convolutional neural networks (CNNs) are effective at learning local morphological features from one-dimensional physiological signals, including characteristic waveform patterns associated with the P wave, QRS complex, and T wave. However, ECG diagnosis also depends on temporal relationships and rhythm patterns extending over multiple cardiac cycles. Recurrent neural networks (RNNs), particularly long short-term memory (LSTM) and gated recurrent unit (GRU) networks, are capable of modeling such temporal dependencies. Combining convolutional and recurrent architectures can therefore provide both morphological and temporal representations of ECG signals.

This work proposes a hybrid deep learning architecture for multi-label classification of 12-lead ECG recordings using the PTB-XL dataset. In the proposed system, raw ECG recordings are first standardized through digital signal processing operations. The recordings are resampled to 200 Hz, band-pass filtered between 0.5 and 40 Hz using a Butterworth filter, normalized independently for each lead, and adjusted to a fixed duration of 10 s. These preprocessing steps reduce unwanted baseline variation and high-frequency noise while providing inputs of consistent dimensions for the neural network.

The proposed model combines a multi-kernel one-dimensional CNN with residual and squeeze-and-excitation

(SE) blocks for feature extraction. Parallel convolutional kernels of different sizes are used to capture ECG characteristics at different temporal scales. Residual connections facilitate the training of deeper feature extraction layers, while SE attention adaptively emphasizes informative feature channels. The extracted features are then processed in parallel by bidirectional LSTM (Bi-LSTM) and bidirectional GRU (Bi-GRU) branches to model temporal dependencies in both forward and backward directions. The representations obtained from the CNN, Bi-LSTM, and Bi-GRU branches are fused and passed to fully connected layers for final classification.

Since a single ECG recording can contain more than one diagnostic label, the problem is formulated as a multi-label classification task rather than a single-class classification problem. Accordingly, independent sigmoid outputs are used for the selected diagnostic categories, and binary cross-entropy with logits is employed as the training loss. The proposed approach is evaluated using metrics including macro area under the receiver operating characteristic curve (AUC), per-label accuracy, exact-match accuracy, precision, recall, F1-score, and class-wise confusion matrices.

The main objective of this work is to develop an end-to-end ECG classification framework that can automatically learn both morphological and temporal characteristics from 12-lead ECG signals. By integrating convolutional feature extraction, channel attention, residual learning, and bidirectional recurrent modeling, the proposed architecture aims to provide a robust approach for automated multi-label ECG abnormality classification.

II. Dataset and Label Definition

The notebook reads 21,837 PTB-XL header files and successfully constructs 21,254 processed examples of shape 12×2000. The 583-record difference reflects records not included in the final cached array; the notebook does not provide a detailed breakdown of exclusion causes. The Challenge collection contains ECGs from several public databases and uses SNOMED-CT diagnosis codes [1]. PTB-XL itself is a large 12-lead ECG dataset [2].

The ten labels are selected by frequency from the available header diagnosis codes. Because labels may co-occur, the task is multi-label rather than mutually exclusive classification.

III. Methods

The proposed system is an end-to-end deep learning framework for multi-label classification of 12-lead electrocardiogram (ECG) recordings. The complete workflow consists of ECG data acquisition, label extraction, signal preprocessing, dataset construction, train-validation-test splitting, hybrid CNN-RNN feature extraction, feature fusion, model training, and performance evaluation. The overall methodology is designed to learn both morphological characteristics and temporal dependencies present in ECG signals.

TABLE I
Selected Diagnosis Labels

No.	Label	Records in processed label matrix
1	Sinus rhythm	18,092
2	Myocardial infarction	5,261
3	Left axis deviation	5,146
4	Abnormal QRS	3,389
5	Left ventricular hypertrophy	2,359
6	T-wave abnormality	2,345
7	Myocardial ischemia	2,175
8	Left anterior fascicular block	1,626
9	Atrial fibrillation	1,514
10	Ventricular ectopics	1,154

A. ECG Data Acquisition and Label Extraction

The proposed system uses ECG recordings from the PTB-XL dataset obtained through the PhysioNet Challenge 2020 data collection. Each recording contains 12 ECG leads along with a corresponding header file. The header provides information such as the number of leads, original sampling frequency, lead gains, and diagnostic codes.

For each recording, the corresponding signal is loaded from the MATLAB file and converted into millivolts using the gain information provided in the header. Diagnostic codes are extracted from the field. Since a single ECG recording may contain multiple diagnostic conditions, the problem is formulated as a multi-label classification task. The most frequent ten diagnostic codes are selected as the target classes. Each ECG is therefore represented by a binary target vector in which a value of one indicates the presence of a particular diagnostic category.

B. ECG Signal Preprocessing

Raw ECG recordings may contain baseline drift, high-frequency noise, variations in sampling frequency, and differences in signal amplitude. Therefore, a digital signal processing pipeline is applied before the signals are provided to the neural network.

First, all recordings are resampled to a common sampling frequency of 200 Hz. This provides a consistent temporal resolution for all ECG signals. The resampling operation is performed using polyphase resampling.

Next, a third-order Butterworth band-pass filter with cutoff frequencies of 0.5 Hz and 40 Hz is applied to each lead. Zero-phase filtering is performed using forward and backward filtering so that important waveform features are not shifted in time.

After filtering, each ECG lead is independently normalized using z-score normalization. For a signal x, the normalized signal is given by

$$x_{norm} = \frac{x - \mu}{\sigma + \epsilon},$$

where μ is the mean of the lead, σ is its standard deviation, and ϵ is a small constant used to avoid division by zero.

Finally, every recording is converted to a fixed duration of 10 s. At a sampling frequency of 200 Hz, each input contains 2000 samples per lead. Signals longer than the required length are truncated, whereas shorter signals are zero-padded. Thus, the final input size of the network is 12×2000.

C. Dataset Construction and Splitting

After preprocessing, recordings that do not contain any of the selected target diagnoses are excluded. The processed ECG signals are stored as input data X, while their corresponding multi-label vectors are stored as Y.

The dataset is divided randomly into three subsets using a fixed random seed of 42. Seventy percent of the recordings are used for training, fifteen percent for validation, and the remaining fifteen percent for testing. The training set is used to optimize the neural network parameters, while the validation set is used to monitor model performance during training and select the best model. The test set is reserved for the final evaluation.

A batch size of 256 is used for training, validation, and testing.

D. Proposed Hybrid CNN-RNN Architecture

The proposed neural network combines convolutional feature extraction with bidirectional recurrent processing. The architecture accepts an input tensor of size 12×2000.

The first stage consists of a multi-kernel one-dimensional CNN. Three parallel convolutional layers with kernel sizes of 5, 15, and 25 are applied to the 12-lead ECG input. Each convolutional layer produces 32 feature channels. The outputs of the three branches are concatenated, producing a 96-channel representation. Different kernel sizes allow the model to capture ECG characteristics at different temporal scales, ranging from narrow waveform changes to broader morphological patterns.

The resulting representation is processed by three residual squeeze-and-excitation blocks. Each residual block contains two one-dimensional convolutional layers, batch normalization, a squeeze-and-excitation attention mechanism, and a shortcut connection. The residual connection allows the input representation to be directly combined with the transformed representation, helping the network learn deeper feature representations.

The squeeze-and-excitation mechanism performs channel-wise attention. Global temporal averaging is first used to obtain a channel descriptor. Two fully connected layers followed by nonlinear activation and sigmoid scaling generate channel weights. These weights are used to emphasize informative feature channels and suppress less useful ones.

The three residual blocks progressively reduce the temporal dimension from 2000 to 1000, 500, and finally 250 samples, while increasing the number of feature channels to 256. A global average over the temporal dimension is then applied to obtain a 256-dimensional CNN feature vector.

E. Bidirectional Recurrent Feature Extraction

The 256-channel feature sequence generated by the convolutional blocks is rearranged into a sequence of 250 time steps with 256 features per step. This sequence is provided to two parallel recurrent branches: a bidirectional long short-term memory (Bi-LSTM) network and a bidirectional gated recurrent unit (Bi-GRU) network.

The Bi-LSTM contains 128 hidden units in each direction and therefore produces a 256-dimensional bidirectional representation. Similarly, the Bi-GRU contains 128 hidden units in each direction and also produces a 256-dimensional representation.

The bidirectional structure enables the network to learn temporal dependencies in both forward and backward directions. Temporal averaging is performed over the recurrent outputs to obtain a fixed-length representation from each branch.

F. Feature Fusion and Classification

Three feature representations are obtained from the proposed architecture: the 256-dimensional CNN representation, the 256-dimensional Bi-LSTM representation, and the 256-dimensional Bi-GRU representation. These representations are concatenated to form a 768-dimensional fused feature vector.

The fused representation is passed through a fully connected classification head consisting of dropout, a linear layer with 256 neurons, a ReLU activation, another dropout layer, and a final linear layer with ten output neurons corresponding to the selected diagnostic classes.

Because the ECG classification problem is multi-label, sigmoid activation is applied independently to the ten output scores during inference. Unlike softmax, sigmoid allows multiple diagnostic categories to be predicted simultaneously for the same ECG recording.

G. Training and Decision Thresholds

The implementation uses binary cross-entropy with logits and positive-class weights derived from label frequencies (square-root ratio capped at four), AdamW with learning rate 10^{-3} and weight decay 10^{-4}, gradient clipping at 1.0, and a validation-AUC-driven learning-rate scheduler. Training was configured for at most 40 epochs with early stopping patience of eight epochs. The recorded run stopped after epoch 25 and achieved its best validation macro AUC of 0.9458 at epoch 17. Per-class thresholds were selected on validation records using bootstrap resampling to optimize F1, then applied once to the held-out test set.

H. Performance Evaluation

The final model is evaluated on the unseen test set. Predicted probabilities are generated using the sigmoid function, and a threshold of 0.5 is used to convert probabilities into binary predictions.

The evaluation includes macro area under the receiver operating characteristic curve (AUC), per-label accuracy, exact-match accuracy, precision, recall, and F1-score. In addition, a multilabel confusion matrix is generated separately for each

TABLE II
OVERALL HELD-OUT TEST RESULTS

Metric	Value
Processed ECG records	21,254
Training / validation / test records	14,880 / 3,172 / 3,202
Trainable parameters	2,116,586
Best validation macro AUC	0.9458
Test macro AUC	0.9430
Mean per-class accuracy	0.9211
Macro F1-score	0.7384

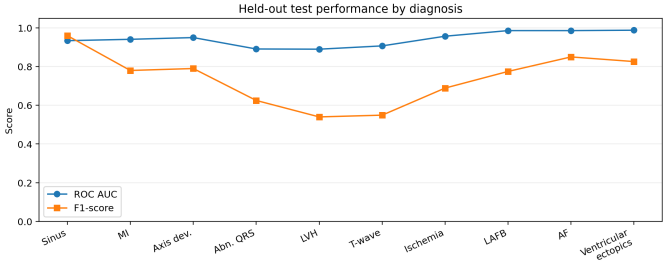


Fig. 1. Per-class ROC AUC and F1-score on the held-out test set.

diagnostic category. Exact-match accuracy is a strict metric because every predicted label for an ECG must match the complete ground-truth label vector for the prediction to be considered correct.

IV. EXPERIMENTAL RESULTS

The test set contains 3,202 records. Since diagnoses can co-occur, performance is calculated independently for each label. Table II reports the aggregate metrics recorded by the notebook, while Table II provides class-wise values. Accuracy should be interpreted alongside precision, recall, F1, and AUC because label prevalence is uneven.

The highest recorded F1-score is 0.960 for sinus rhythm. Atrial fibrillation and ventricular ectopics have F1-scores of 0.850 and 0.826, respectively. Lower F1-scores for left ventricular hypertrophy (0.540) and T-wave abnormality (0.549) indicate that these labels remain challenging at the selected thresholds. Macro AUC is higher than macro F1, which is consistent with ranking ability being stronger than thresholded classification for some labels.

V. CONFUSION MATRIX ANALYSIS

Figure 2 presents the confusion matrices for the ten ECG diagnosis labels evaluated on the held-out test set. Each matrix shows the number of correctly and incorrectly classified records, where the rows represent the true labels and the columns represent the predicted labels. The diagonal entries indicate correct predictions, while the off-diagonal entries represent classification errors.

The model performs well in identifying sinus rhythm, with 2,660 true-positive and 319 true-negative predictions. Atrial fibrillation achieves 195 true positives, with 32 false negatives and 37 false positives. Ventricular ectopics achieves 152 true

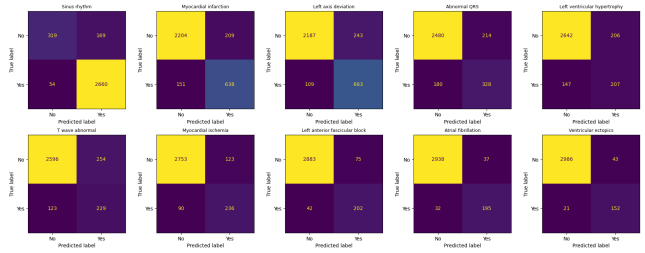


Fig. 2. Confusion matrices for the ten ECG diagnosis labels on the held-out test set.

positives, with 21 false negatives and 43 false positives. These results indicate good identification performance for these two arrhythmia labels.

For myocardial infarction, the model correctly identifies 638 positive records, while 151 positive records are incorrectly classified as negative. Left axis deviation achieves 663 true positives, with 109 false negatives. In comparison, abnormal QRS has 328 true positives and 180 false negatives. Left ventricular hypertrophy and T-wave abnormality show 207 and 229 true positives, respectively, indicating that these classes remain more challenging for the model.

Overall, the confusion matrices demonstrate that classification performance varies across diagnosis labels. The differences in false-positive and false-negative counts highlight the importance of evaluating each class separately in this multi-label ECG classification task, rather than relying only on aggregate metrics.

VI. DISCUSSION

The results support the feasibility of combining multi-scale convolutional features with bidirectional recurrent features for multi-label ECG classification. The multi-kernel stem offers filters with different temporal receptive fields; residual connections support deeper feature learning; SE blocks reweight channel features; and LSTM/GRU branches provide complementary sequence representations.

Important limitations remain. First, the experiment uses the PTB-XL directory from the Challenge package and should not be described as training on every database in the full Challenge collection. Second, label counts are imbalanced and some diagnoses have relatively few positive test examples. Third, the split is record-level but not reported as patient-level; if multiple recordings from one patient are present, patient-level leakage cannot be ruled out from the notebook alone. Fourth, thresholds are tuned on a validation set and are specific to this data split. Fifth, results from a single seed do not quantify run-to-run variation. Finally, the code and metrics are research outputs and are not a substitute for clinician review or external clinical validation.

VII. CONCLUSION

This paper presented a hybrid CNN–LSTM–GRU pipeline for multi-label classification of ten frequent diagnoses from 12-lead ECGs. The executed notebook processed 21,254

records and obtained test macro AUC 0.9430 and macro F1 0.7384. Performance varies by diagnosis, with lower F1 values for some structurally and morphologically related labels. Future work should include patient-level splitting, repeated-seed evaluation, calibration analysis, external-dataset testing, comparison with CNN-only and recurrent-only baselines, and clinician-led error analysis.

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This project has been a valuable learning experience, help ing us strengthen our technical knowledge, problem-solving abilities, and understanding of machine learning applications in biomedical signal analysis.

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